Virtual Mitosis Lab **Name** ______________________

Find the virtual lab at:
http://www.biology.arizona.edu/cell_bio/activities/cell_cycle/cell_cycle.html

Answer the questions below as you read the first two pages of the virtual lab website.

1. Why are root tips so useful for observing mitosis? ______________________

Many root-tip cells are actively dividing (high mitotic index).

2 DNA replicates in S phase, making sister chromatids.

Prophase. No.

3 Nuclear membrane breaks down.

They align to ensure equal chromosome distribution.

4 Each daughter cell gets one copy of each chromosome.

Main anaphase trait: sister chromatids separate and move apart.

They are pulled toward opposite poles by spindle fibers.

5 Telophase: nuclei reform and chromosomes decondense.

Cytokinesis begins and the cell starts pinching in.

6 ______________________

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➢ Click NEXT at the bottom of the page. You do not need to copy the data table onto separate paper, as it's included here.

➢ Now you will begin the lab. You will be presented with 36 onion root tip cells. Determine which phase each cell is in. (You will know if you are correct or not.) Place a tally mark in the appropriate column in the table below. You should have a total of 36 tally marks when you are finished.

Interphase	Prophase	Metaphase	Anaphase	Telophase

➢ Complete the data table below by following the instructions for each row.

Data Table

	Interphase	Prophase	Metaphase	Anaphase	Telophase	Totals
# of cells	20	10	3	2	1	36
% of cells	55.56%	27.78%	8.33%	5.56%	2.78%	100%
# of minutes	588.9	294.4	88.3	58.9	29.4	1060

You'll need a calculator to complete some of the rows in this table.

- To fill in the **"# of cells"** row, count the number of cells in each of the phases from the data table on page 1 of this lab and write the number in the appropriate column.

- To fill in the **"% of cells"** row, divide the number of cells in each phase by 36 and then multiply by 100. (We use 36 because that's the total number of cells you observed.

- To fill in the **"# of minutes"** row you'll determine how long a cell spends in each phase. To to this, multiply the % of cells by 1060, which is the total number of minutes in an onion root tip cell cycle.

- You can check your work for each row by adding up the numbers in each column. The total should be exact in the # of cells row, and extremely close, if not exact, in the other two rows.

Analysis Questions

Interphase had the most cells because cells spend most of the cycle there.

Interphase: intact nucleus and diffuse chromatin.

Prophase: condensed chromosomes become visible.

Nuclear membrane begins to break down in prophase.

3. Describe how the cells in metaphase looked compared to the cells in anaphase.

Metaphase: chromosomes line up at center.

Anaphase: sister chromatids separate to opposite poles.

Daughter cells are genetically identical to the parent cell.

They keep the same chromosome number as the parent cell.

5. Label the illustration below using the following terms:

Centromere

Chromatid

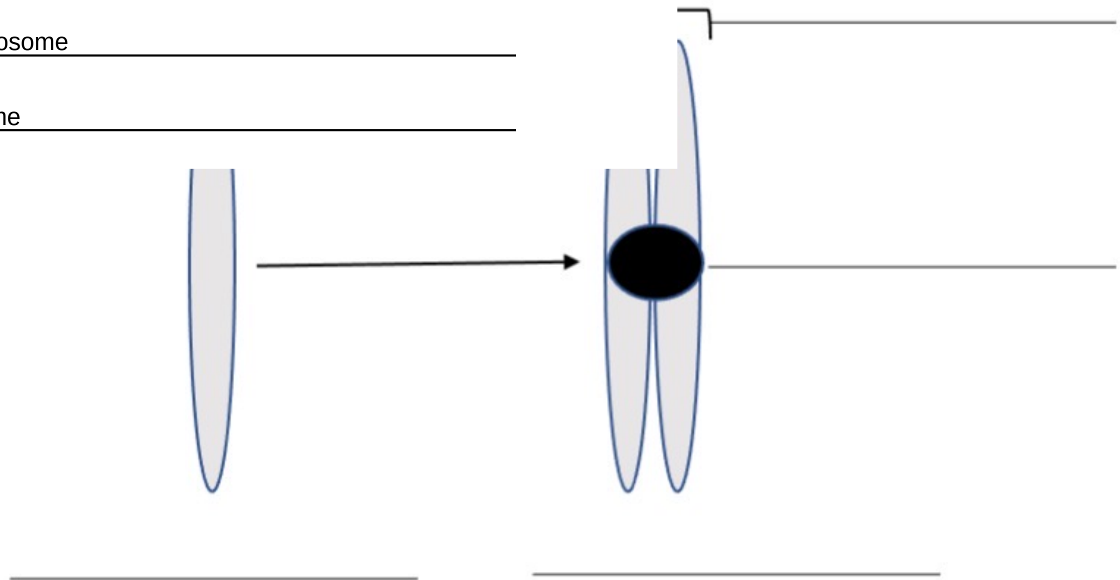
Sister chromatids

Replicated chromosome

Single chromosome

s, replicated chromosome,

ne



DNA replication in S phase of interphase.

Once.

Two daughter cells.

Cytokinesis (cleavage furrow formation).

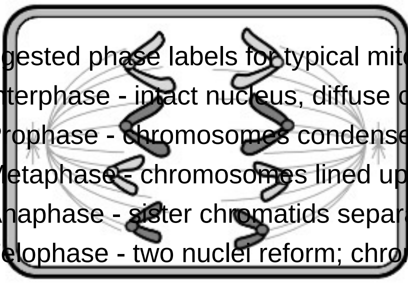
Interphase

(S phase)

Label the phase each cell represents. Then describe the characteristics of each cell that helped you decide.

Suggested phase labels for typical mitosis c
1) Interphase - intact nucleus, diffuse chrom
2) Prophase - chromosomes condense and
3) Metaphase - chromosomes lined up acros
4) Anaphase - sister chromatids separate ar
5) Telophase - two nuclei reform; chromoso

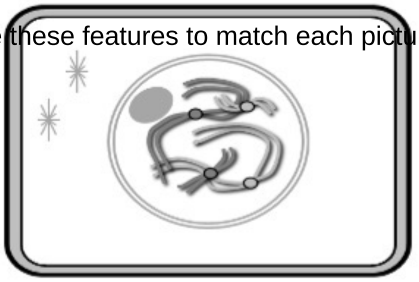
Use these features to match each pictured c



A.

A. Metaphase

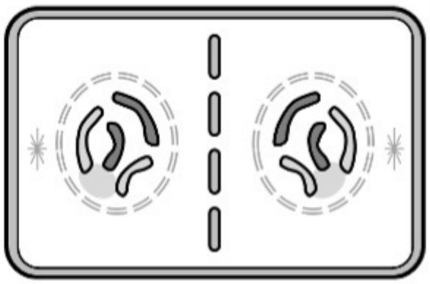
Chromosomes aligned at the center.



B.

B. Interphase

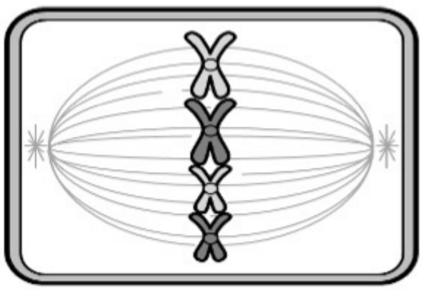
Intact nucleus; diffuse chromatin.



C.

C. Telophase/Cytokinesis

Two nuclei reform; membrane pinches in.



D.

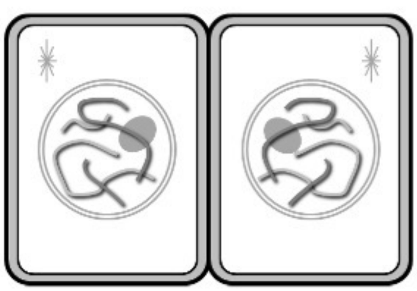
D. Metaphase

Chromosomes line up across middle.

E.

E. Prophase

Chromosomes condense; nucleus breaks down.



F.

F. Telophase

Chromosomes decondense; daughter nuclei form.